# GATE- ALL BRANCHES

## ENGINEERING MATHEMATICS

**Linear Algebra**

**Lecture No. 7**

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# Recap of previous lecture

$\rightarrow A_{m\times n}\vec{x}_{n\times 1} = \vec{b}_{m\times 1}$.

$\rho(A) = \rho(A|\vec{b}) = n \rightarrow$ unique solution.

$\rho(A) = \rho(A|\vec{b}) < n \rightarrow$ Infinitely Many Solutions.

$\rho(A) \neq \rho(A|\vec{b}) \rightarrow$ No Solution.

$\rightarrow$ Gauss Elimination.

$\rightarrow$ LU Decomposition.

$A = LU$

For $\vec{b} = \vec{0}$;

$$\underbrace{A\vec{x} = \vec{0}}_{\text{Homogeneous system}}$$

$\vec{x} = \vec{0}$ (Trivial solution)

For $A_{m\times n}\vec{x}_{n\times 1} = \vec{0}_{m\times 1}$

to have Non-tivial solution

$$\rho(A) < \min\{m, n\}$$

For $A_{n\times n}\vec{x}_{n\times 1} = \vec{0}_{n\times 1}$

$\rho(A) < n \Rightarrow \det(A) = 0$.

# Topics to be Covered

**Topic** Eigen Values and Eigen Vectors

**Topic** Diagonalization

## MCQ

Let r and s be real numbers. If $A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 0 & 3 \\ r & s & 0 \end{pmatrix}$ and $b = \begin{pmatrix} 1 \\ 1 \\ s-1 \end{pmatrix}$, then the system of linear equations AX = b has

A. ~~no~~ solutions for s≠2r.

B. infinitely many solutions for s = 2r ≠ 2.

C. a unique solution for s = 2r = 2.

D. infinitely many solutions for s = 2r =2.

Sol: Given System is $A\vec{x} = \vec{b}$;

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 0 & 3 \\ \lambda & S & 0 \end{pmatrix}; \quad \vec{b} = \begin{pmatrix} 1 \\ 1 \\ S-1 \end{pmatrix}.$$

$$\Rightarrow (A|\vec{b}) = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 1 \\ \lambda & S & 0 & S-1 \end{pmatrix} \xrightarrow[R_3 \to R_3 - \lambda R_1]{R_2 \to R_2 - 2R_1} (A|\vec{b}) \sim \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & -4 & 3 & -1 \\ 0 & \underbrace{\boxed{S-2\lambda}}_{=0} & 0 & \underbrace{S-1-\lambda}_{\neq 0} \end{pmatrix}$$

For $\underbrace{S - 2\lambda = 0}$ $\Rightarrow \rho(A) = 2$

$\rightarrow$ For Infinitely Many Solutions; $\rho(A|\vec{b}) = 2 \Rightarrow S-1-\lambda = 0$

$\Rightarrow S - \lambda = 1$

$\Rightarrow \boxed{\lambda = 1 \Rightarrow S = 2}$

$\Rightarrow$ For infinitely Many solutions;

$\underbrace{S = 2\lambda = 2}$.

$\rightarrow$ For No-solution: $S = 2\lambda$; $S-1-\lambda \neq 0$

$\Rightarrow S = 2\lambda \neq 2$

If $S \neq 2\lambda$.

$\Rightarrow \rho(A) = 3$. unique Solution

Ex: If $S - 2\lambda = 1$

$R_3 \to 4R_3 + R_2$

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & -4 & 3 & -1 \\ 0 & 0 & 3 & \bigcirc \end{pmatrix}$$

# Eigen Values and Eigen vectors

For a Matrix $A_{n\times n}$; a Vector $\vec{x}$ $(\vec{x} \neq \vec{0})$ is Said to be an eigen vector of 'A' corresponding to an eigen Value '$\lambda$' if

$$\boxed{A\vec{x} = \lambda\vec{x}}$$

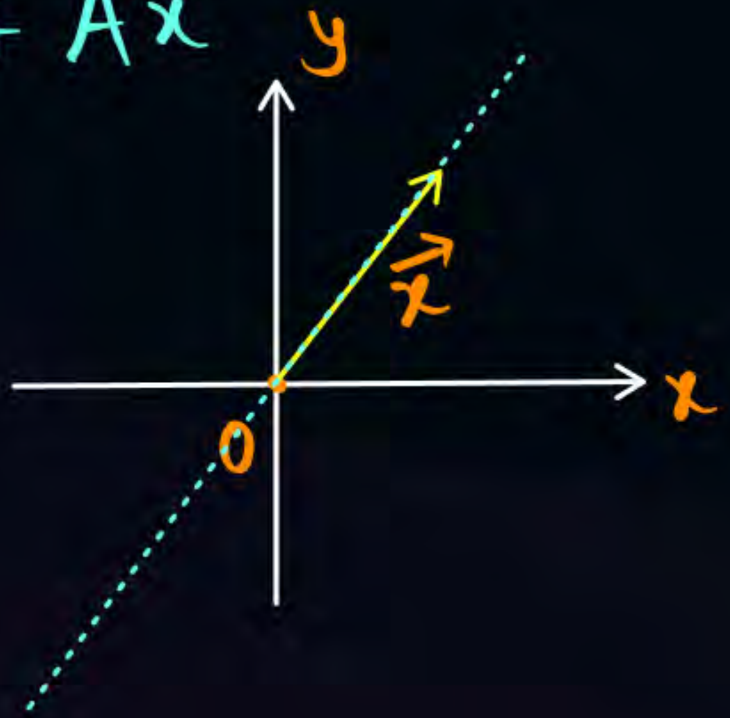
Scaling factor

$(\lambda, \vec{x})$ is called Eigen Pair.

For $\lambda$ being a scalar;

direction (line of action) of $A\vec{x}$ is Same as that of $\vec{x}$.

$A\vec{x} = \vec{y}$



$$A\vec{x} = \lambda\vec{x}.$$

$$\Rightarrow A\vec{x} - \lambda I\vec{x} = \vec{0}.$$

$$\Rightarrow \boxed{(A - \lambda I)\vec{x} = \vec{0}.}$$

Homogeneous system of linear Equations.

For eigen vectors to exist (i.e $\vec{x} \neq \vec{0}$);

the Above system should have Non-tivial solution

$$\Rightarrow \rho(A - \lambda I) < n.$$

$$\Rightarrow \boxed{\det(A - \lambda I) = 0}$$

Characteristic Equation.

To calculate eigen values of '$A$'$_{n\times n}$ we solve the Equation $\underbrace{|A - \lambda I| = 0}$.

Polynomial Equation.

# Polynomials

If $a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n = 0$. then

① If $a_0 + a_1 + a_2 + \dots + a_n = 0$ then $x = 1$ is a root.

② If $a_n = 0$, then $x = 0$ is a root.

③ If Sum of Coefficients of odd Powers of 'x' is equal to Sum of Coefficients of even Powers of x, then $x = -1$ is a root.

Ex: $x^3 - 7x^2 + 4x^1 + 12x^0 = 0 \longrightarrow -1 - 7 - 4 + 12 = 0$ ✓

Sum of coeff. of odd Powers of $x = 1 + 4 = 5$

Sum of coeff. of even Powers of $x = -7 + 12 = 5$

$\Rightarrow x = -1$ is a root.

→ If $\lambda_1, \lambda_2, \lambda_3, \ldots, \lambda_n$ are roots of Polynomial Equation $f(x) = 0$; then $\lambda_1 + k, \lambda_2 + k, \ldots, \lambda_n + k$ (where 'k' is a Scalar), are roots of the equation $f(x-k) = 0$;

<u>Ex:</u> Consider $x^2 - 7x + 12 = 0$

$\Rightarrow x = 3, 4$

$\Rightarrow$ Let $x_1 = 3 - 2 = 1$ ✓

$x_2 = 4 - 2 = 2$ ✓

$\Rightarrow x^2 - 3x + 2 = 0$

Decrease roots by 2;

then Polynomial is $f(x+2) = 0$

$\Rightarrow (x+2)^2 - 7(x+2) + 12 = 0$

$\Rightarrow x^2 - 3x + 2 = 0$

$$|A-\lambda I|=0.$$

For 'A' being an 'n×n' Matrix; $|A-\lambda I|=0$ is a Polynomial equation of degree 'n'.

Ex: Say $A=\begin{bmatrix}1&2&1\\2&1&1\\1&1&2\end{bmatrix}$

Consider $|A-\lambda I|=\begin{vmatrix}1-\lambda&2&1\\2&1-\lambda&1\\1&1&2-\lambda\end{vmatrix}=0$

$$\Rightarrow (1-\lambda)\{(1-\lambda)(2-\lambda)-1\}-2\{4-2\lambda-1\}+1\{2-(1-\lambda)\}=0$$

$$\Rightarrow (1-\lambda)\{(\lambda^2-3\lambda+1)\}-2\{3-2\lambda\}+(1+\lambda)=0$$

$$\Rightarrow \lambda^2-3\lambda+1-\lambda^3+3\lambda^2-\lambda-6+4\lambda+1+\lambda=0 \Rightarrow \boxed{\lambda^3-4\lambda^2-\lambda+4=0}$$

(OR)

$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}_{3\times 3}$ ; Characteristic Polynomial $C_A(\lambda)$ is

$$\boxed{C_A(\lambda) = \lambda^3 - tr(A).\lambda^2 + \{M_{11} + M_{22} + M_{33}\}\lambda - \det(A) = 0.}$$

$M_{ij} \rightarrow$ Minor of element $a_{ij}$

$\therefore C_A(\lambda) = \lambda^3 - 4\lambda^2 + (1 + 1 + (-3))\lambda - (-4) = 0$

$\Rightarrow \lambda^3 - 4\lambda^2 - \lambda + 4 = 0$

On solving, $\underbrace{\lambda = 1 \text{ is a root}}$. $\Rightarrow$ $\lambda_1 + \lambda_2 + \lambda_3 = tr(A) = 4 \Rightarrow \lambda_2 + \lambda_3 = 3$

$\lambda_1 \lambda_2 \lambda_3 = \det(A) = -4$; $\lambda_2 \cdot \lambda_3 = -4$

$\Rightarrow 1, 4, -1$ are eigen Values

Since $A_{3\times 3}$ has characteristic Polynomial

$$C_A(\lambda) = \lambda^3 - tr(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - \det(A) = 0.$$

$\therefore$ The roots (Eigen Values) are $\lambda_1, \lambda_2, \lambda_3$.

$\Rightarrow \lambda_1 + \lambda_2 + \lambda_3 = tr(A)$

$\Rightarrow$ $\lambda_1 + \lambda_2 + \lambda_3 = tr(A) =$ Sum of Principal diagonal elements of 'A'

$$\lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1 = M_{11} + M_{22} + M_{33}.$$

$$\lambda_1\lambda_2\lambda_3 = \det(A)$$

$ax^3 + bx^2 + cx + d = 0$;

$\alpha, \beta, \gamma$.

$\alpha + \beta + \gamma = -b/a$

$\alpha\beta + \beta\gamma + \gamma\alpha = c/a$.

$\alpha\beta\gamma = -d/a$.

→ Ex:  $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \Rightarrow C_A(\lambda) = \lambda^3 - 4\lambda^2 - \lambda + 4 = 0.$

Eigen Values of 'A' are 1, 4, −1.

Sum, $\lambda_1 + \lambda_2 + \lambda_3 = 1 + 4 - 1 = tr(A)$

$\underbrace{\lambda_1 \cdot \lambda_2 + \lambda_2 \cdot \lambda_3 + \lambda_3 \cdot \lambda_1} = (1 \times 4) + (4 \times -1) + (1 \times -1)$

$= \cancel{4} - \cancel{4} - 1 = -1. = \overset{M_{11}}{\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix}} + \overset{M_{22}}{\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix}} + \overset{M_{33}}{\begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix}}$

$= 1 + 1 + (-3) = -1.$

$\lambda_1 \lambda_2 \lambda_3 = 1(4)(-1) = -4$

$= \det(A).$

**MCQ**

Let A be a 3×3 matrix such that $A^2=A$. Then it is necessary that

(A) A is the identity matrix or the zero matrix.

(B) the determinant of $A^4$ is either 0 or 1 .

(C) the rank of A is 3 .

(D) A has one imaginary eigenvalue. ✗

Sol: Given $A_{3\times3}$ is $\underbrace{A^2=A}_{\text{Idempotent Matrix.}}$

$A=\begin{bmatrix}4 & -1\\ 12 & -3\end{bmatrix} \neq \underbrace{I \text{ (or) } O}_{\text{Even } 3\times3 \,/\!.}$

$A^2=A \Rightarrow |A|^2=|A|$

$\Rightarrow \underbrace{|A|=0}$ (or) 1.

$\left.\begin{array}{l}\Rightarrow |A|^4=0^4=0\\ \quad\text{(OR)}\\ |A|^4=1^4=1\end{array}\right\} \Rightarrow |A|^4=0 \text{ (or) } 1.$

It has Either 2 $\underbrace{\text{Imaginary}}_{}$ / 0 imaginary roots.

$\underbrace{\lambda_1, \lambda_2, \lambda_3}_{} \longrightarrow 3$ roots.

## MCQ

Considering the matrix $\begin{pmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 0 \end{pmatrix}$

Which one of the following statements is INCORRECT?

- (A) One of its eigenvalues is zero ✓
- (B) It has two purely imaginary eigenvalues. ✓
- (C) It has a non-zero real eigenvalue. ✗
- (D) The sum of its eigenvalues is zero. ✓

Sol: $C_A(\lambda) = \lambda^3 - 0\cdot\lambda^2 + (9+4+1)\lambda - 0 = 0$

$\Rightarrow \lambda^3 + 14\lambda = 0$

$\Rightarrow \lambda\{\lambda^2 + 14\} = 0$

$\boxed{\Rightarrow \lambda = 0;\ \lambda = \sqrt{14}\, i;\ \lambda = -\sqrt{14}\, i}$

$z = \cancel{a}^{0} + ib \rightarrow$ Purely imaginary

## Properties of Eigen Values

→ If $\lambda_1, \lambda_2, \lambda_3, \ldots \lambda_n$ are eigen values of $A_{n\times n}$ ; then

① $\lambda_1 + \lambda_2 + \cdots + \lambda_n = \sum_{i=1}^{n} \lambda_i = tr(A)$

② $\lambda_1 \cdot \lambda_2 \cdots \lambda_n = \prod_{i=1}^{n} \lambda_i = \det(A)$.

③ Eigen Values of $A^{\alpha}$ are $\lambda_1^{\alpha}, \lambda_2^{\alpha}, \ldots, \lambda_n^{\alpha}$

④ Eigen Values of $A^{-1}$ are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \ldots, \frac{1}{\lambda_n}$ $(\lambda_1 \neq 0; \lambda_2 \neq 0, \ldots \lambda_n \neq 0)$

⑤ Eigen Values of adj(A) are $\frac{|A|}{\lambda_1}, \frac{|A|}{\lambda_2}, \ldots, \frac{|A|}{\lambda_n}$. $(\lambda_1 \neq 0; \lambda_2 \neq 0; \ldots \lambda_n \neq 0)$

⑥ For a Scalar 'k'; Eigen Values of $A + kI$ are $\lambda_1 + k, \lambda_2 + k, \ldots, \lambda_n + k$

⑦ Eigen Values of $aA^2 + bA + c.I$ are $a\lambda_1^2 + b\lambda_1 + c;\ a\lambda_2^2 + b\lambda_2 + c;\ \ldots\ a\lambda_n^2 + b\lambda_n + c$

⑧ → If a Matrix is Upper Triangular / lower Triangular / Diagonal Matrix, then the eigen Values are the Principal diagonal elements of that Matrix.

Ex. $A = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \Rightarrow |A - \lambda I| = \begin{vmatrix} a_{11}-\lambda & 0 & 0 \\ a_{21} & a_{22}-\lambda & 0 \\ a_{31} & a_{32} & a_{33}-\lambda \end{vmatrix} = 0$

$\Rightarrow (a_{11} - \lambda)(a_{22} - \lambda)(a_{33} - \lambda) = 0 \Rightarrow \lambda = a_{11}, a_{22}, a_{33}$

→ zero is definitely an Eigen Value of an odd-order skew symmetric Matrix.

i.e For $A_{n\times n} = -A^T_{n\times n}$ when 'n' is odd;

$\lambda = 0$ is definitely an Eigen Value.

→ The Eigen Values of a real symmetric Matrix are always real.

i.e For $A = A^T$; If $A\vec{x} = \lambda\vec{x}$ then $\lambda \in \mathbb{R}$.

→ The Eigen Values of a Skew-symmetric Matrix are either zero (or) Purely Imaginary. i.e Eigen Values are $0, +ia; -ia$ & so on.

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Home Work.

If 1, 0, and –1 are the eigenvalues of a $3 \times 3$ matrix A, then the trace of $A^2 + 5A$ is equal to

________________.

NAT

Home Work.

#Q Let $I$ denote the identity matrix of order 7 , and A be a 7×7 real matrix having characteristic polynomial $C_A(\lambda) = \lambda^2 (\lambda - 1)^\alpha (\lambda + 2)^\beta$, where $\alpha$ and $\beta$ are positive integers. If A is diagonalizable and rank (A) rank (A+ 2$I$), then rank (A-I) is________ (in integer).

# 2 mins Summary

$\rightarrow A\vec{x} = \lambda \vec{x}$

$|A - \lambda I| = 0$

$\downarrow$

Calculating Eigen values

$\downarrow$

Properties of eigen values.

# THANK - YOU